



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**FINAL EXAMINATION
SEMESTER III, SESSION 2020/2021
PART II**

COURSE CODE	:	FSPC 0034
COURSE NAME	:	CHEMISTRY II
PROGRAMME	:	ASASI UTMSPACE
DURATION	:	1 HOUR 30 MINUTES
DATE	:	15 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. THERE ARE **FOUR (4)** QUESTIONS. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (8) printed pages including this page.

QUESTION 5 (10 MARKS)

a) Two (2) cell notations are given as follows:

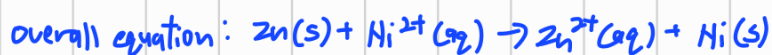
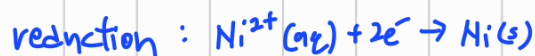
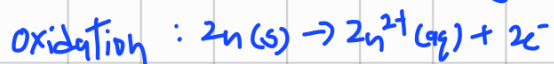
Cell Notation	$\text{Zn(s)}/\text{Zn}^{2+}(\text{aq}) \parallel \text{Ni}^{2+}(\text{aq})/\text{Ni(s)}$	$\text{Ni(s)}/\text{Ni}^{2+}(\text{aq}) \parallel \text{Ag}^{+}(\text{aq})/\text{Ag(s)}$
$E^{\circ}_{\text{cell}}/\text{Volt}$	0.51	1.05

Based on the information in the table,

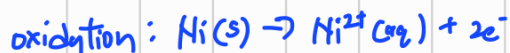
- i. Write the chemical equations to represent each of the galvanic cell.
- ii. Given the standard electrode potential of $\text{Ni}^{2+}(\text{aq})/\text{Ni(s)}$ is 0.23 V, determine the standard electrode potential for $\text{Zn(s)}/\text{Zn}^{2+}(\text{aq}) \parallel \text{Ag}^{+}(\text{aq})/\text{Ag(s)}$.
-  iii. If certain amount of water is added to the silver half-cell in (ii) and the voltmeter reading dropped to 1.48 V, determine the new concentration of the $\text{Ag} \mid \text{Ag}^{+}(\text{aq})$.

(2 + 6 + 2 marks)

5) a) (i) First galvanic cell oil rig



Second galvanic cell



$$\text{Cii) } E^{\circ}_{\text{cell}} = E^{\circ}_{\text{reduction}} - E^{\circ}_{\text{oxidation}}$$

$$0.51 = -0.23 - (E^{\circ}_{\text{Zn}})$$

$$E^{\circ}_{\text{Zn}} = -0.74 \text{ V}$$

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{reduction}} - E^{\circ}_{\text{oxidation}}$$

$$1.05 = E^{\circ}_{\text{Ag}} - (-0.23)$$

$$E^{\circ}_{\text{Ag}} = +0.82 \text{ V}$$

$$E^{\circ}_{\text{cell}} = E^{\circ}_{\text{red}} - E^{\circ}_{\text{oxi}}$$

$$= +0.82 - (-0.74)$$

$$= 1.56 \text{ V}$$

$$\text{Ciii) } 1.48 = 1.56 - \frac{0.0592}{2} \log\left(\frac{[\text{Ag}^{+}]}{1}\right)$$

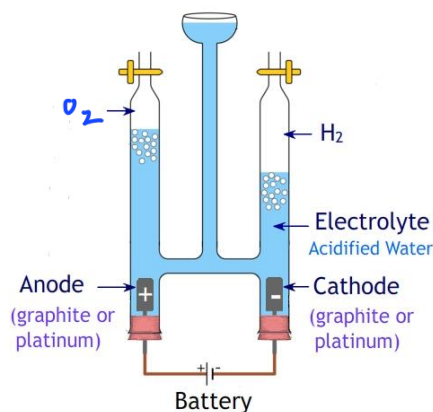
$$0.08 = 0.0296 \log[\text{Ag}^{+}]$$

$$2.7027 = \log[\text{Ag}^{+}]$$

$$[\text{Ag}^{+}] = 504.31 \text{ M}$$

QUESTION 6 (10 MARKS)

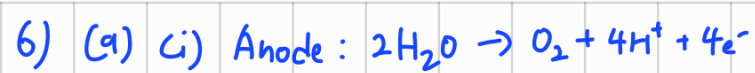
- a) An electrolysis cell as shown below was used to generate hydrogen gas at room temperature. A current of 2.35A passes through the acidified water solution for 40 minutes.



- i. Write the half reactions at the anode and cathode.
 - ii. Determine the volume of the hydrogen collected if the pressure is exactly 1 atm. Assume the reaction occurs at STP conditions.
- (2 + 4 marks)
- b) Electrolysis was performed for several electrolytes. How much of each of the following metals can be discharged when the same quantity of electricity is passed through these electrolytes?

- i. Silver nitrate, AgNO_3 .
- ii. Copper (II) sulfate, CuSO_4 .
- iii. Identify which Law is applicable for the above and state the Law.

(1 + 1 + 2 marks)



(ii) $m_{\text{H}_2} = \frac{2(2.35)(2400)}{4(96500)}$

$= 0.0292 \text{ g}$

$\text{mol of H}_2 = \frac{0.0292}{2}$

$= 0.0146 \text{ mol}$

Volume = 0.0146×22.4

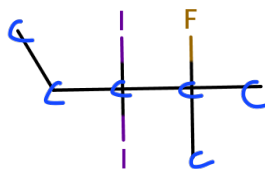
$= 0.327 \text{ dm}^3$



(iii) Faraday Second Law

QUESTION 7 (10 MARKS)

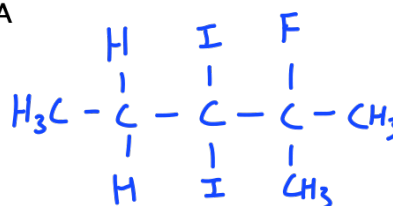
- a) The secondary carbon in structure A has been successfully converted to tertiary carbon after undergone a few chemical reactions.



Structure A

For this new structure, you are required to:

- i. Draw its NEW structural formula.



- ii. Provide its IUPAC name.

3,3-diiodo-2-chloro-2-methyl-pentane

- iii. State its functional group

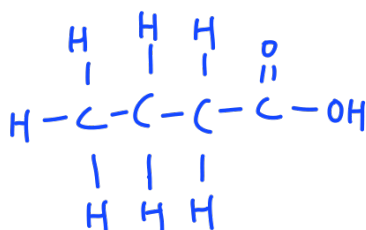
alkyl group

(1 + 2 + 1 marks)

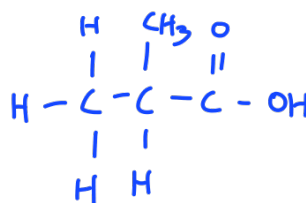
- b) Structural isomers, also known as constitutional isomers are common in organic compounds. Draw **three (3)** structural isomers of esters with formula $C_4H_8O_2$. State the name of each structural isomer.

C_3H_7COOH

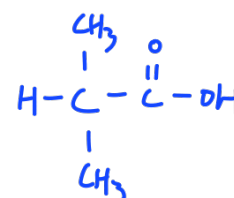
(6 marks)



1-butanoic acid



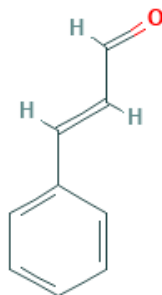
2-methyl-1-propanoic acid



isopropyl ethanoic acid

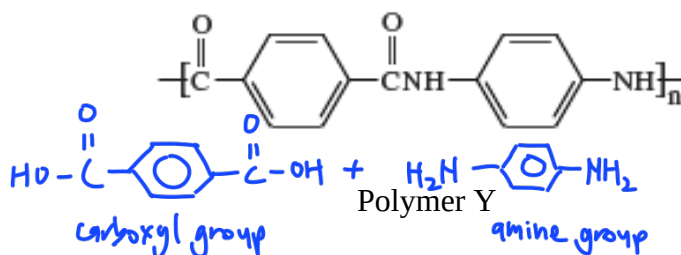
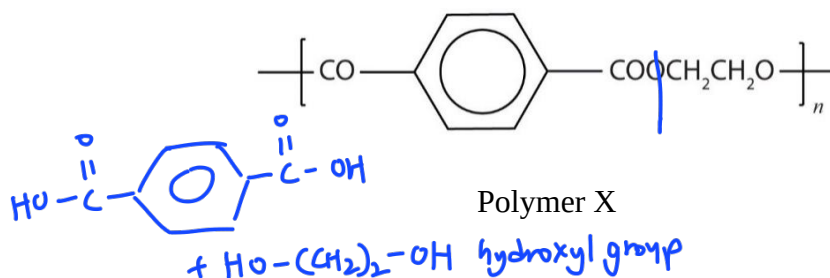
QUESTION 8 (10 MARKS)

- a) Cinnamaldehyde ($\text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$) is an organic compound with a yellow liquid appearance. It gives cinnamon flavour, odour, and has a sweet taste. It has a role as a hypoglycaemic agent, antifungal agent and as a flavouring agent. Explain **ONE (1)** experimental procedure involved in testing cinnamaldehyde and the expected results.



(4 marks)

- b) Polymerization is widely used in the textile, paint and aesthetic industries. The repeating units of **TWO (2)** polymers are shown below:



- Write the structural formulae of the monomers used to make polymers X and Y.
- Explain one (1) difference between X and Y in terms of their structure.
hydroxyl group + carboxyl / amine group + carboxyl group
- Identify the type of polymerization used to make Y. Justify your answer.

- Condensation polymerization

- OH⁻ from

(2 + 2 + 2 marks)

Appendix 2 : LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Appendix 3 : STANDARD REDUCTION POTENTIAL (E°_{Red})

$\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$	+1.69
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.51
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au}(\text{s})$	+1.50
$\text{ClO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 8\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.39
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow 2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\ell)$	+1.33
$2\text{HNO}_3(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow \text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\ell)$	+1.30
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\ell)$	+1.23
$\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$	+1.22
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{Br}^-(\text{aq})$	+1.07
$\text{Hg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Hg}(\ell)$	+0.85
$\text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{Cl}^-(\text{aq}) + 2\text{OH}^-(\text{aq})$	+0.84
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow \text{Ag}(\text{s})$	+0.80
$\text{NO}_3^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow \text{NO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$	+0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{O}_2(\ell)$	+0.70
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow 2\text{I}^-(\text{aq})$	+0.54
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 4\text{e}^- \rightarrow 4\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$	+0.34
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\ell)$	+0.17
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}^{2+}(\text{aq})$	+0.15
$\text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{S}(\text{aq})$	+0.14
$\text{AgBr}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{Br}^-(\text{aq})$	+0.07
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_{2(\text{g})}$	0.00
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Sn}(\text{s})$	-0.14
$\text{AgI}(\text{s}) + \text{e}^- \rightarrow \text{Ag}(\text{s}) + \text{I}^-(\text{aq})$	-0.15
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ni}(\text{s})$	-0.26
$\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Co}(\text{s})$	-0.28
$\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow \text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$	-0.36
$\text{Se}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2\text{Se}(\text{aq})$	-0.40
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow \text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Fe}(\text{s})$	-0.45
$\text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{e}^- \rightarrow \text{NO}(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.46
$\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow 2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$	-0.69
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cr}(\text{s})$	-0.91
$\text{Se}(\text{s}) + 2\text{e}^- \rightarrow \text{Se}^{2-}(\text{aq})$	-0.92
$\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow \text{SO}_3^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$	-0.93
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow \text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ca}(\text{s})$	-2.87
$\text{Ba}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Ba}(\text{s})$	-2.91
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow \text{Li}(\text{s})$	-3.04